



Sri Lanka Institute of Information Technology

**IT3021 - Data warehousing and
Business Intelligence**

Assignment 02

It23728226

ASHFAN GM

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1. Introduction

This assignment is based on a hospital healthcare dataset that contains information about patient visits, treatments, and outcomes. The dataset includes details such as patient demographics, medical conditions, admission and discharge dates, length of stay, total cost, and satisfaction levels.

The data was organized into a data warehouse using a star schema to support analytical processing. In this assignment, the data warehouse is used to build an OLAP cube using SSAS, perform analysis using Excel, and create visualizations using Power BI for decision-making.

1.1 ER Diagram (Source System)

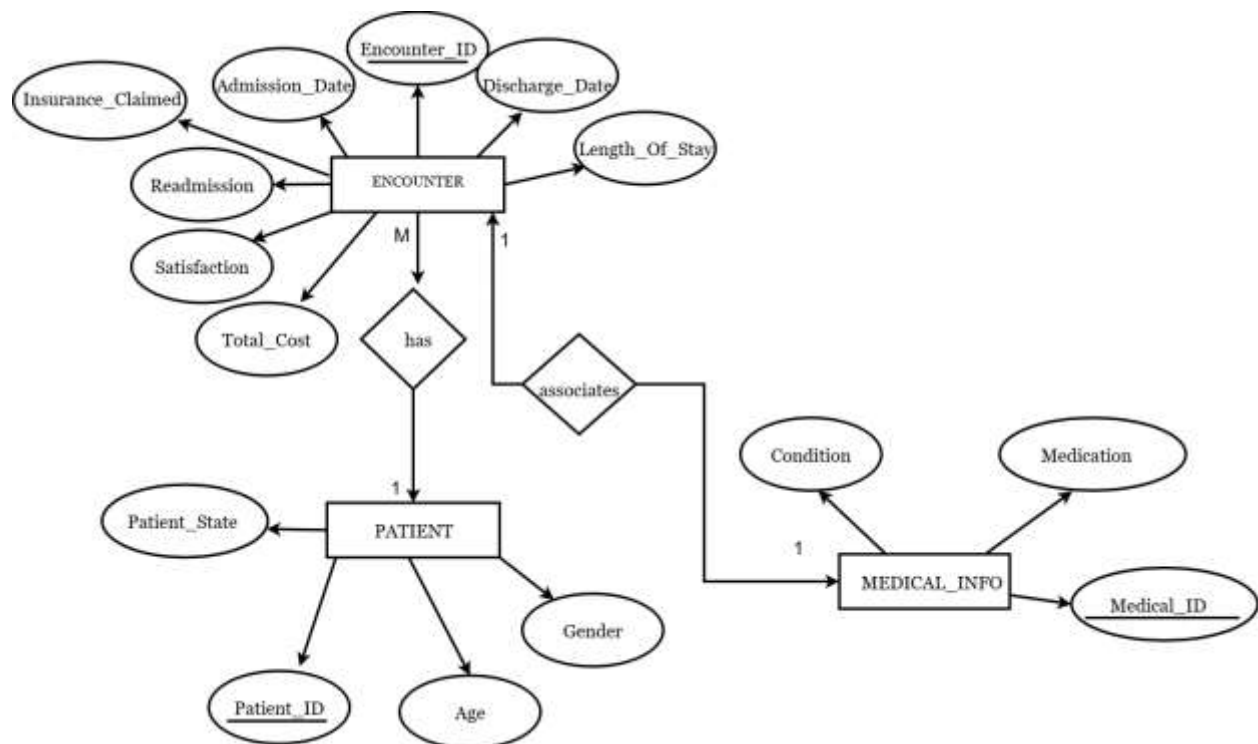


Figure 1- er diagram

The ER diagram represents the structure of the original operational database used as the data source. It includes entities such as Encounter, Patient, and Medical Information along with their relationships. This structure is normalized and supports transactional data processing.

1.2 Star Schema Design

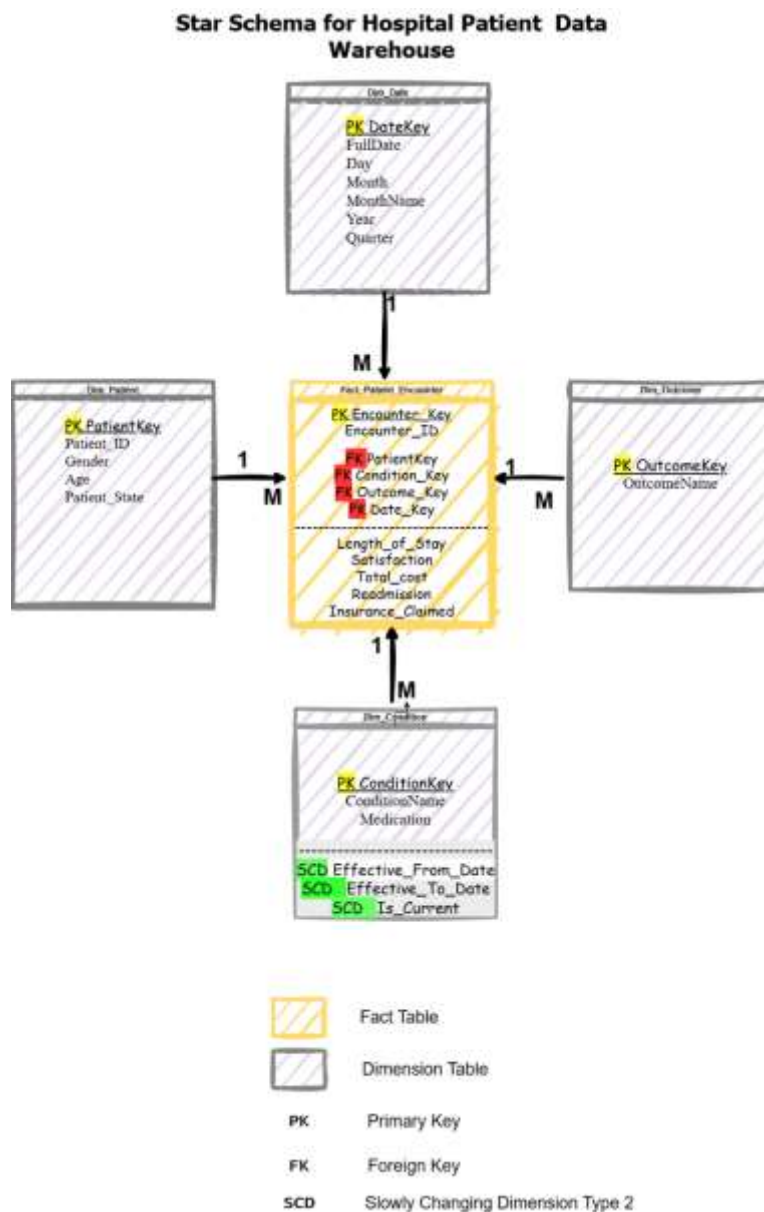
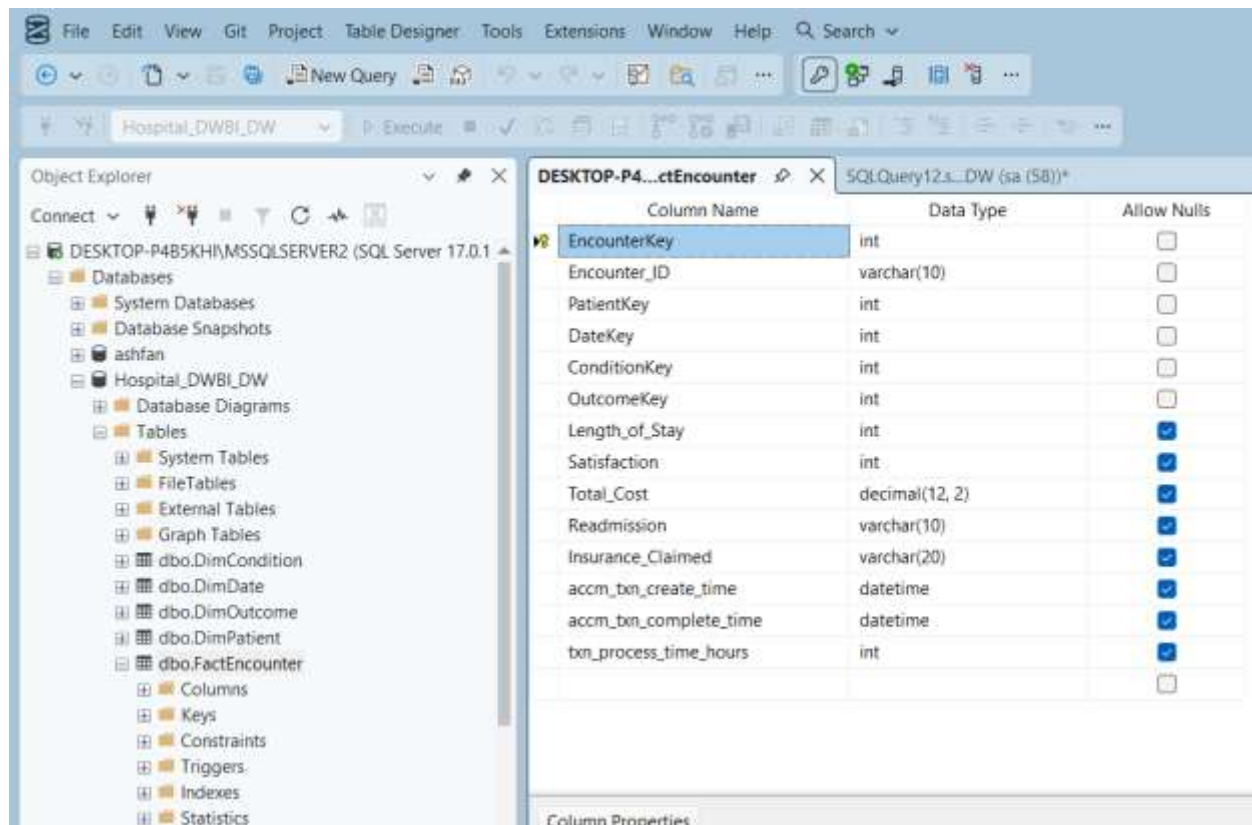


Figure 2 - star schema

The data warehouse follows a star schema design where a central fact table is connected to multiple dimension tables. This structure simplifies analytical queries and improves performance. The fact table stores measurable data, while dimension tables provide descriptive attributes for analysis.

1.3 Fact Table — FactEncounter



The screenshot displays the SQL Server Enterprise Designer interface. The Object Explorer on the left shows the database structure for 'Hospital_DWBI_DW', including dimension tables like DimCondition, DimDate, DimOutcome, and DimPatient, and the central fact table FactEncounter. The main pane shows the table design for 'FactEncounter', listing columns, their data types, and nullability.

Column Name	Data Type	Allow Nulls
EncounterKey	int	☐
Encounter_ID	varchar(10)	☐
PatientKey	int	☐
DateKey	int	☐
ConditionKey	int	☐
OutcomeKey	int	☐
Length_of_Stay	int	☒
Satisfaction	int	☒
Total_Cost	decimal(12, 2)	☒
Readmission	varchar(10)	☒
Insurance_Claimed	varchar(20)	☒
accm_txn_create_time	datetime	☒
accm_txn_complete_time	datetime	☒
txn_process_time_hours	int	☒

Figure 3 - fact table design

The FactEncounter table is the central table in the data warehouse. It contains measurable data related to hospital visits such as length of stay, total cost, satisfaction level, and readmission status. It also includes foreign keys linking to dimension tables such as Patient, Date, Condition, and Outcome.

Additionally, timestamps and transaction processing time are stored for analysis.

1.4 Dimension Tables

The dimension tables provide descriptive information to support analysis:

- **DimPatient:** Stores patient-related details such as age, gender, and state.
- **DimDate:** Contains date attributes including day, month, quarter, and year for time-based analysis.
- **DimCondition:** Stores medical condition details and includes Slowly Changing Dimension (SCD Type 2) attributes to track historical changes.
- **DimOutcome:** Stores outcome information of patient visits.

Step 2: SSAS Cube implementation

1. Create SSAS Project

A new Analysis Services Multidimensional Project was created in Visual Studio to develop the OLAP cube. This project is used to design and deploy multidimensional models for data analysis.

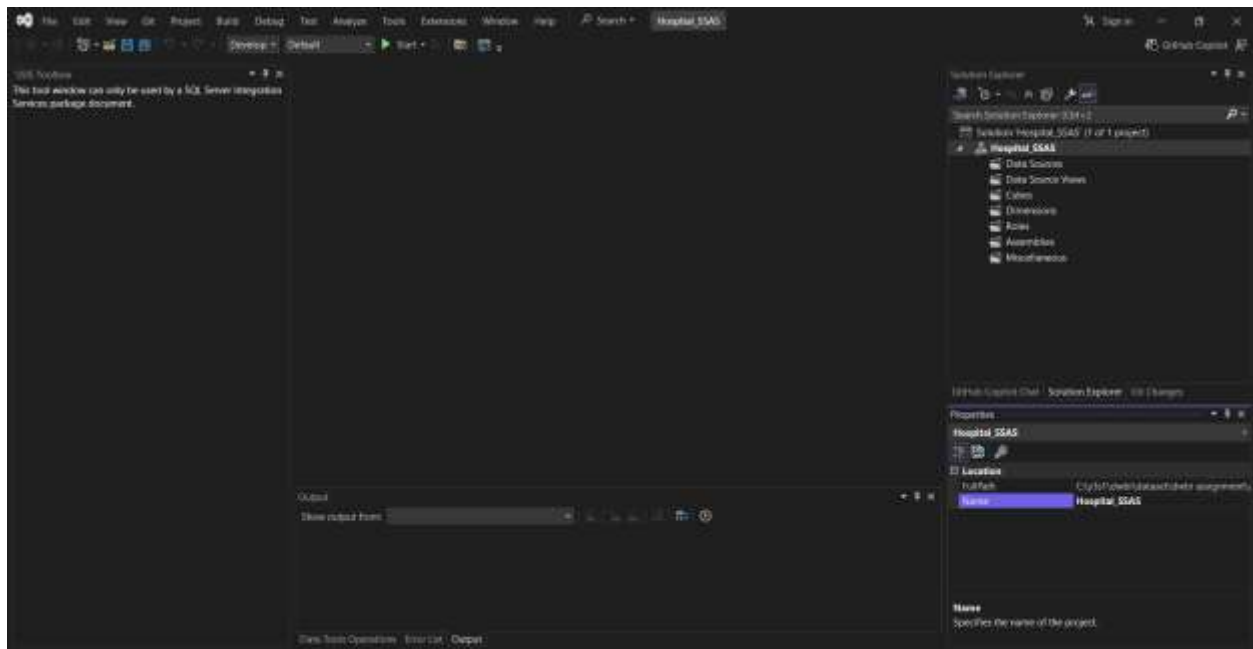


Figure 4 - create SSAS project

2. Create New Data Source

A data source was created to establish a connection between the SSAS project and the Hospital_DW database. This connection allows the cube to access the data stored in the data warehouse

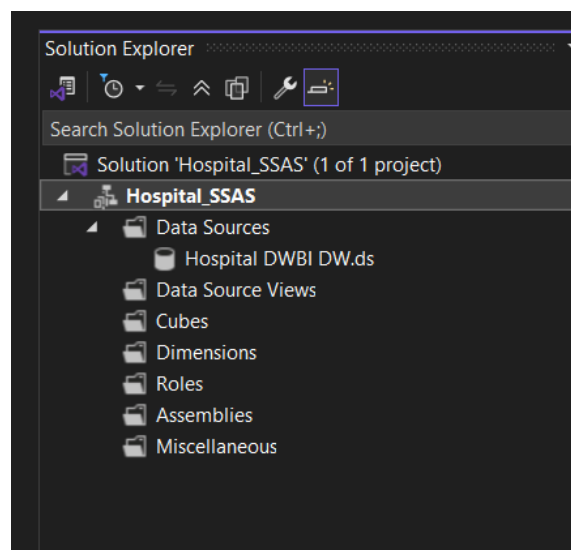


Figure 5 - data source

3. Create Data Source View

A Data Source View was created by selecting the fact table and related dimension tables from the data warehouse. This view visually represents the relationships between tables and serves as the foundation for building the OLAP cube.

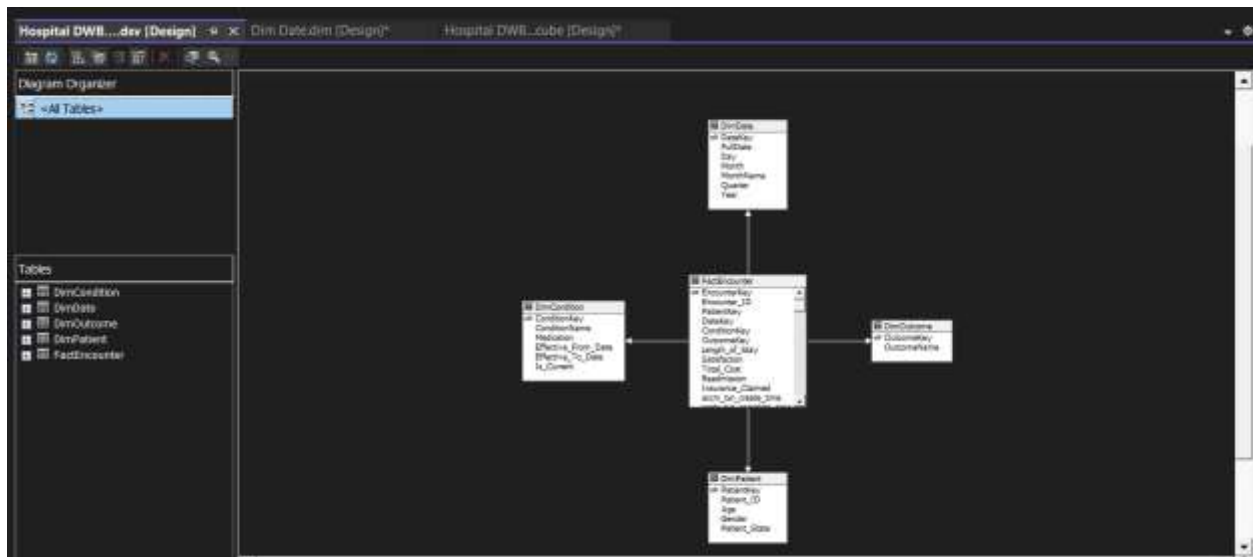


Figure 7- data source views

4. Cube Creation

A cube was created using the FactEncounter table as the measure group. Relevant measures such as total cost, satisfaction, length of stay, and transaction processing time were selected. Dimension tables including Patient, Date, Condition, and Outcome were added to enable multidimensional analysis.

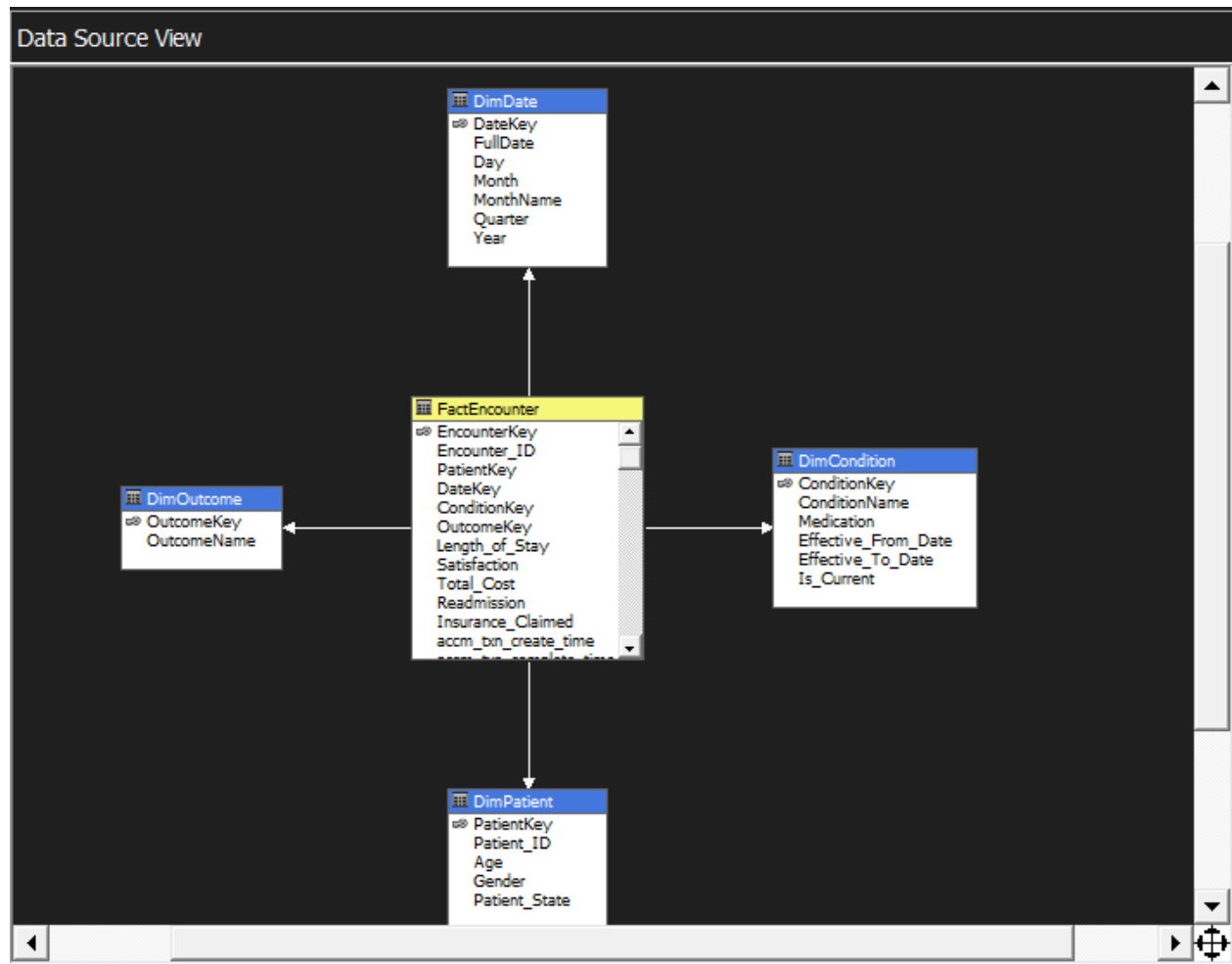


Figure 8- cube creation

5. Create Hierarchy

The cube structure was configured by organizing measures and dimensions. A hierarchy was created in the Date dimension (Year → Quarter → Month → Day) to support OLAP operations such as drill-down and roll-up. This improves the analytical capability of the cube.

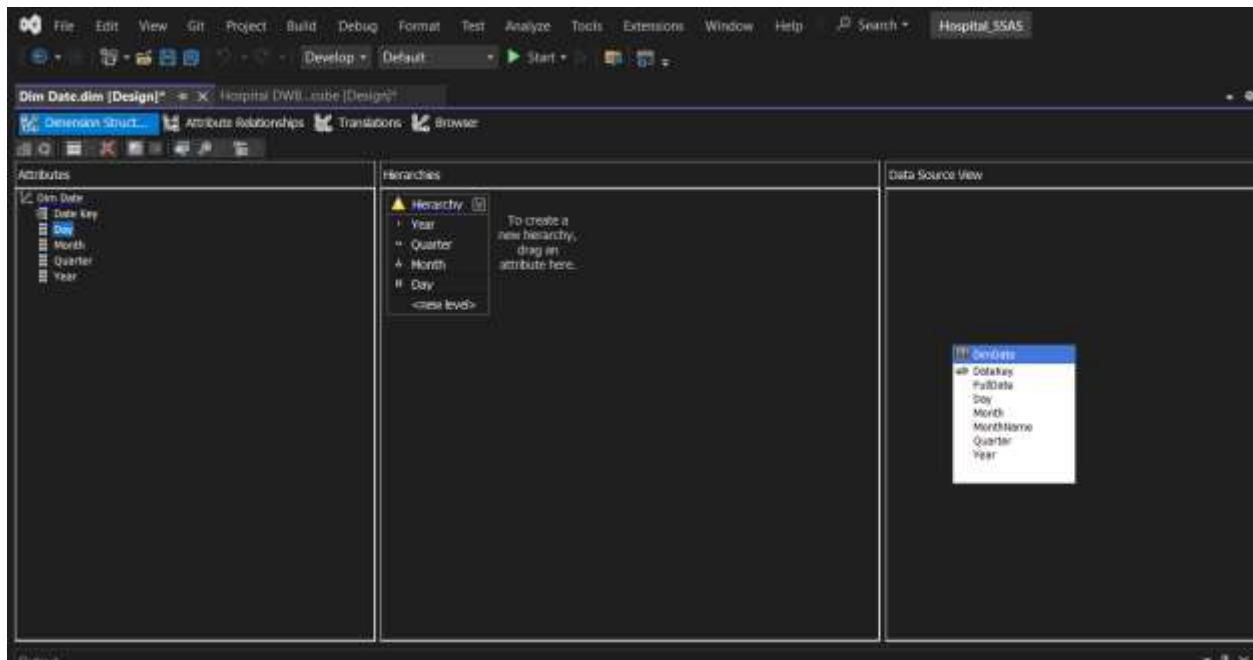


Figure 9 -date hierarchy

6. Cube Deployment and Processing

The cube was deployed to SQL Server Analysis Services (SSAS) using Visual Studio. After deployment, the cube was processed to load data from the data warehouse into the cube. All dimensions and the fact table were successfully processed without errors. This confirms that the cube is correctly configured and ready for analytical operations.

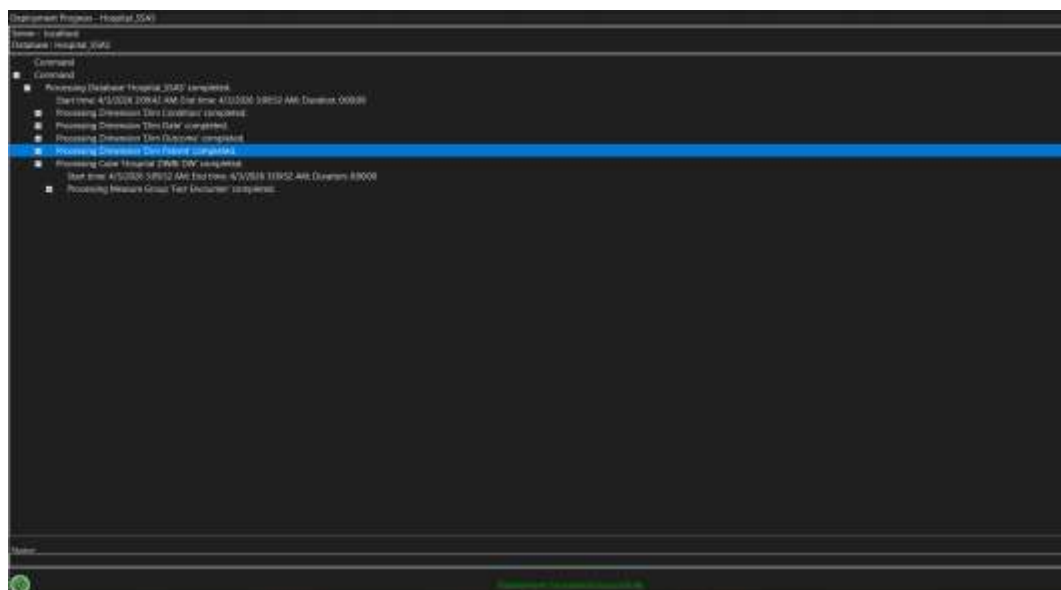


Figure 10- deployment of cube

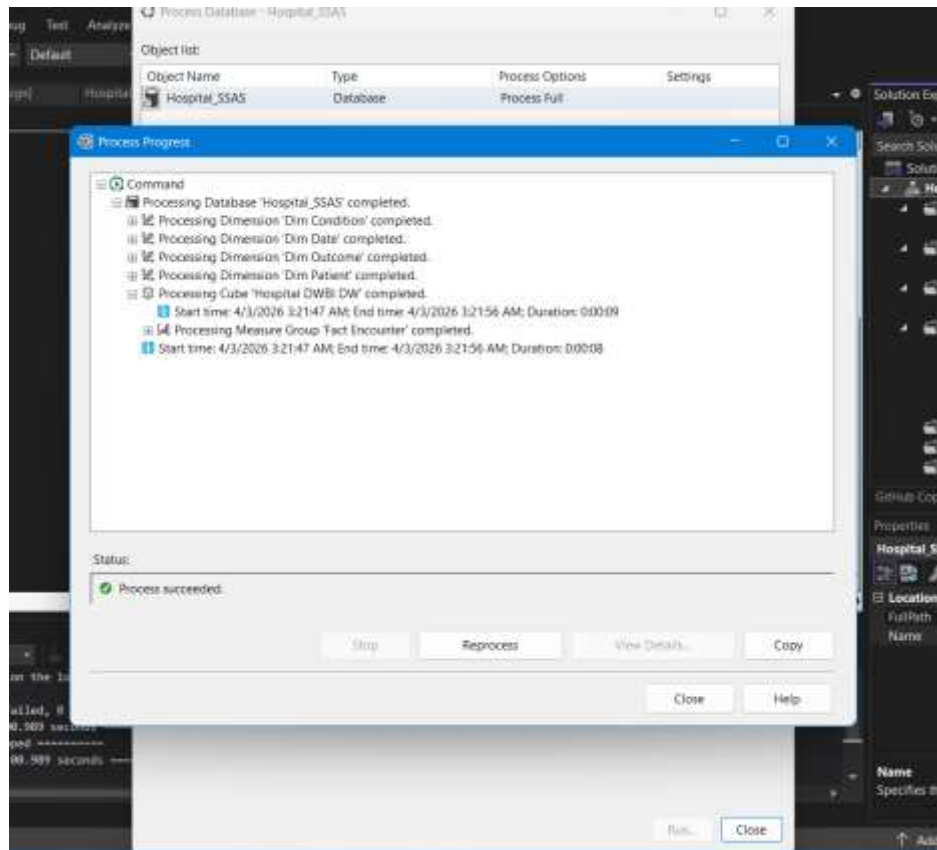


Figure 11 - cube processing

Step 3: Demonstration of OLAP operations

3.1 Connecting Excel to SSAS Cube

An Excel workbook was connected to the SSAS cube using the Data tab. The Analysis Services server was specified, and the Hospital_SSAS cube was selected.

3.2 Created pivot table

Created PivotTable – Generated a PivotTable report in Excel linked to the Hospital_DW cube for OLAP analysis, selecting Totalcost as the primary measure.

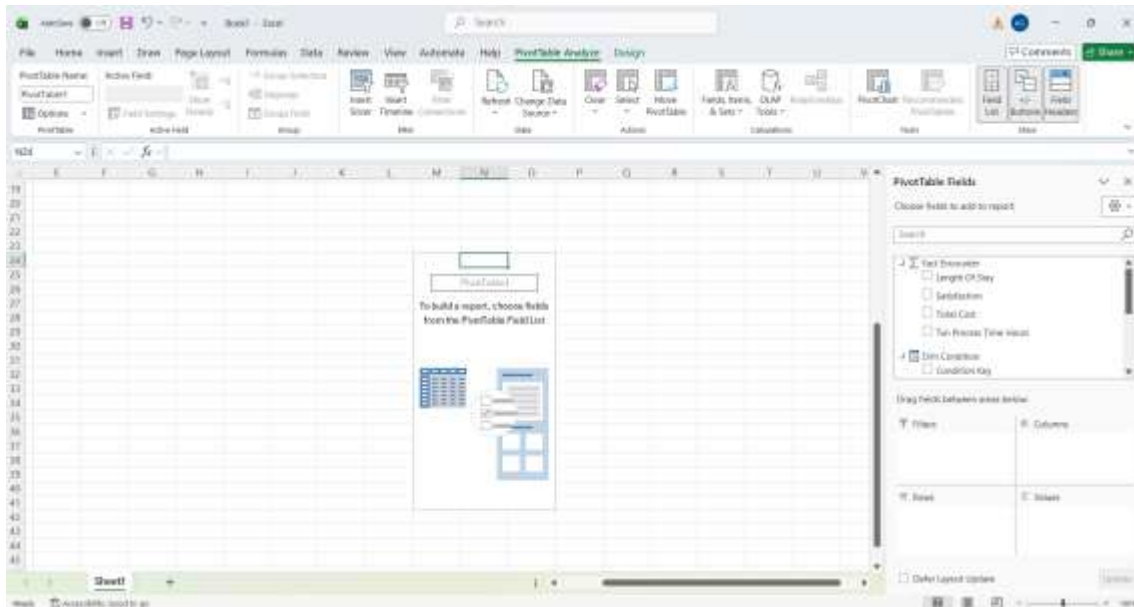


Figure 12 – pivot table

3.3 ROLL-UP OPERATION

Roll-up operation was performed by aggregating the Total Cost measure at the Year level using the Date dimension. This provided a summarized view of data, allowing analysis of overall yearly performance.

Year Labels	Total Cost
2022	44360878.72
2023	46293428.26
2024	61129458.46
2025	42137505.85
Grand Total	191060080.3

Figure 13 – roll up operation

3.4 DRILL-DOWN OPERATION

Drill-down operation was performed by expanding the Date hierarchy from Year to Quarter, Month, and Day levels. This allowed detailed analysis of data at different levels of granularity.

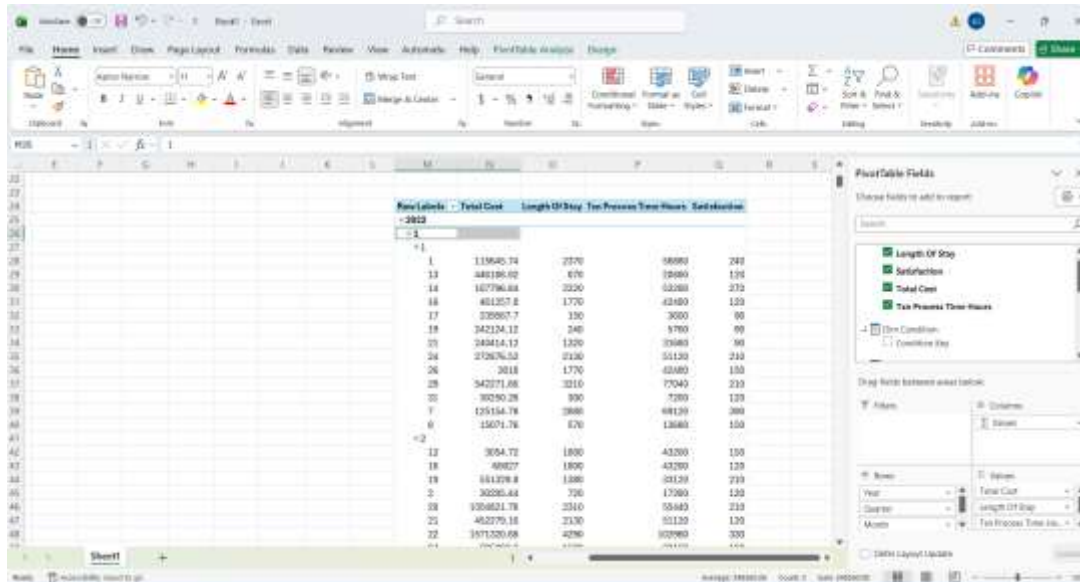
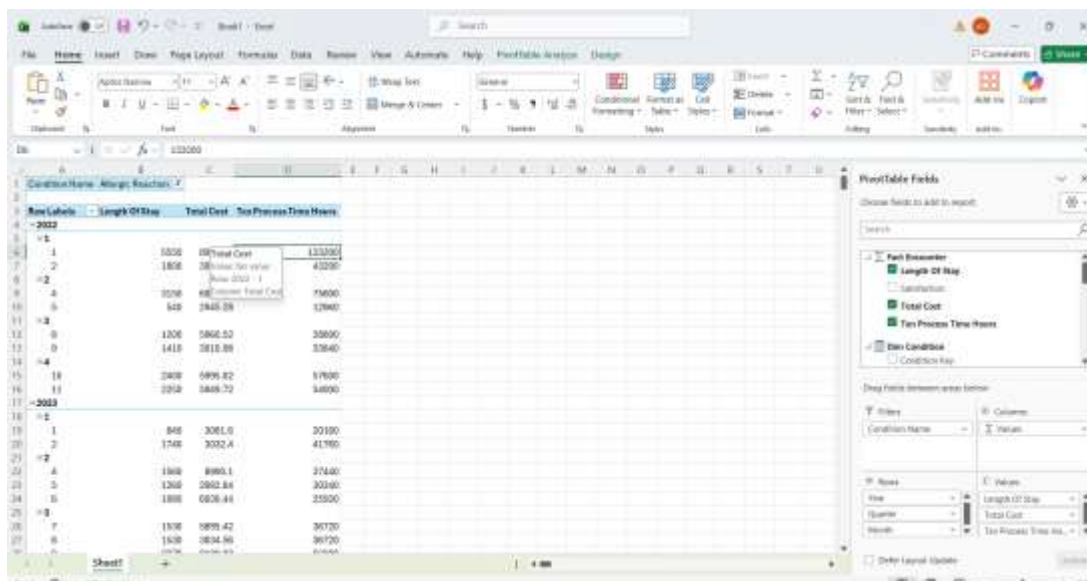


Figure 14 -drill down operation

3.5 SLICE OPERATION

Slice operation was performed by filtering the data based on a single dimension, such as Condition. This allowed focused analysis of data for a specific category.



3.6 DICE OPERATION

Dice operation was performed by applying multiple filters such as Condition and Gender. This enabled multi-dimensional analysis by selecting a subset of data based on multiple criteria.

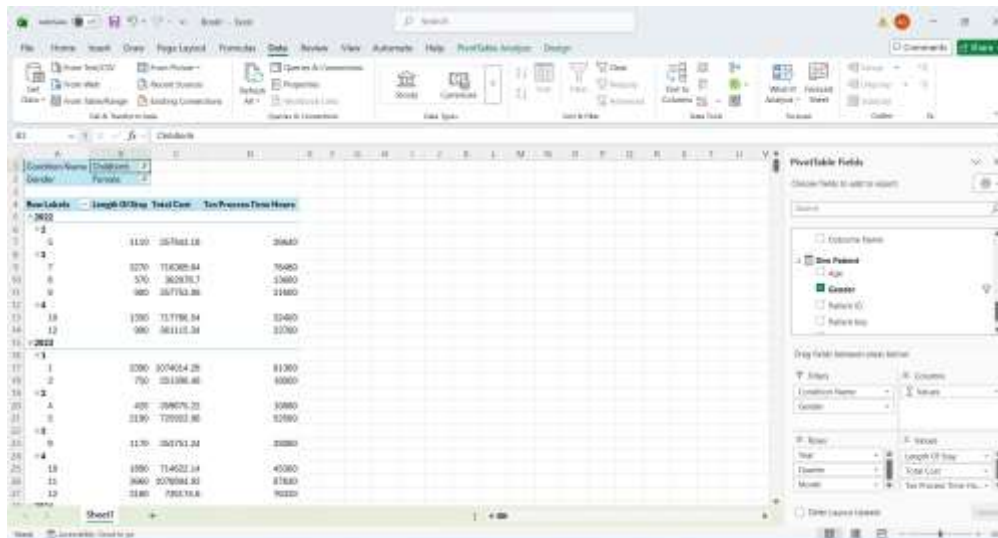


Figure 15-dice operation

3.7 PIVOT OPERATION

Pivot operation was performed by changing the orientation of dimensions between rows and columns. This provided a different perspective of the data and improved analysis flexibility.



Figure 16 -pivot operation

Step 4: PowerBI Reports

4.1 Data Connection and Preparation

Power BI Desktop was connected to the SSAS cube using a live connection. This allowed real-time access to the processed multidimensional data without requiring additional data transformation.

The cube structure (FactEncounter and dimensions such as Date, Patient, Condition, and Outcome) was directly available in Power BI, enabling efficient report creation.

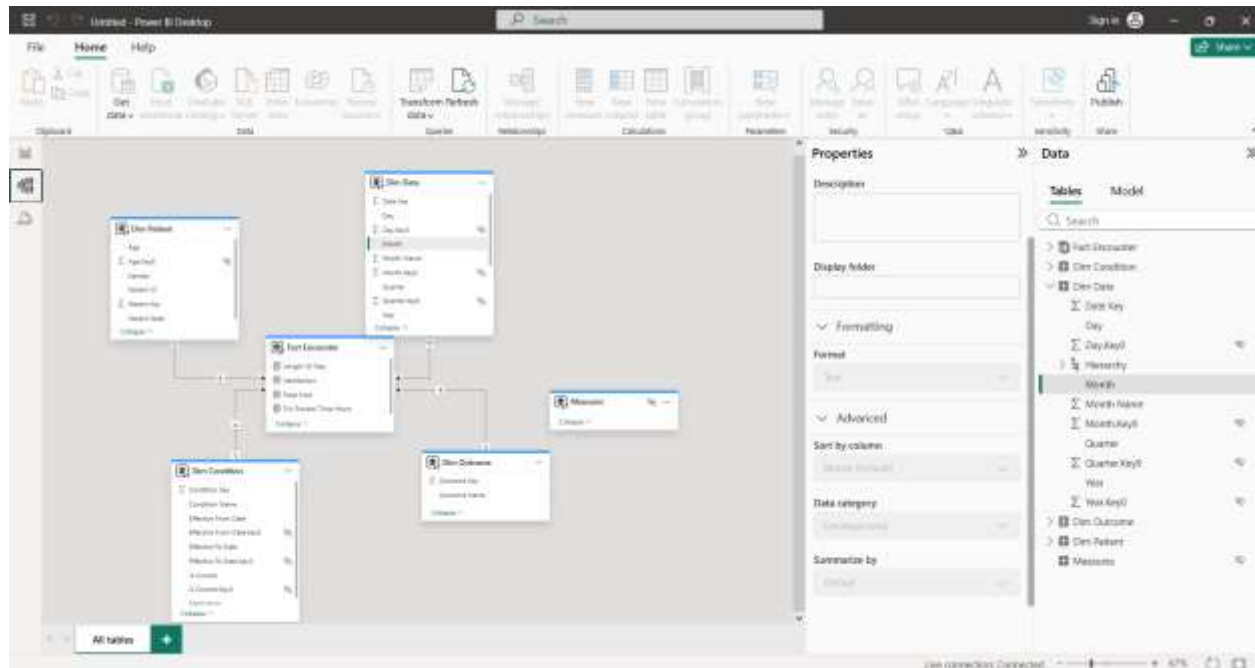


Figure 17– power BI data connection

DAX Usage

- No calculated columns/measures created
- Measures (Total Cost, Length of Stay, etc.) were reused from cube

Since a live SSAS connection was used, DAX creation was not required in Power BI

4.2 Report 1 — Matrix Visual

A Matrix visual was created to display detailed tabular data. It shows measures across different dimensions such as time and medical conditions. This helps compare values and identify patterns.

Steps -

1. Insert **Matrix visual**
2. Add fields:
 - Rows → Condition Name, Year
 - Columns → Month
 - Values → Total Cost, Length of Stay
3. Adjust layout and formatting

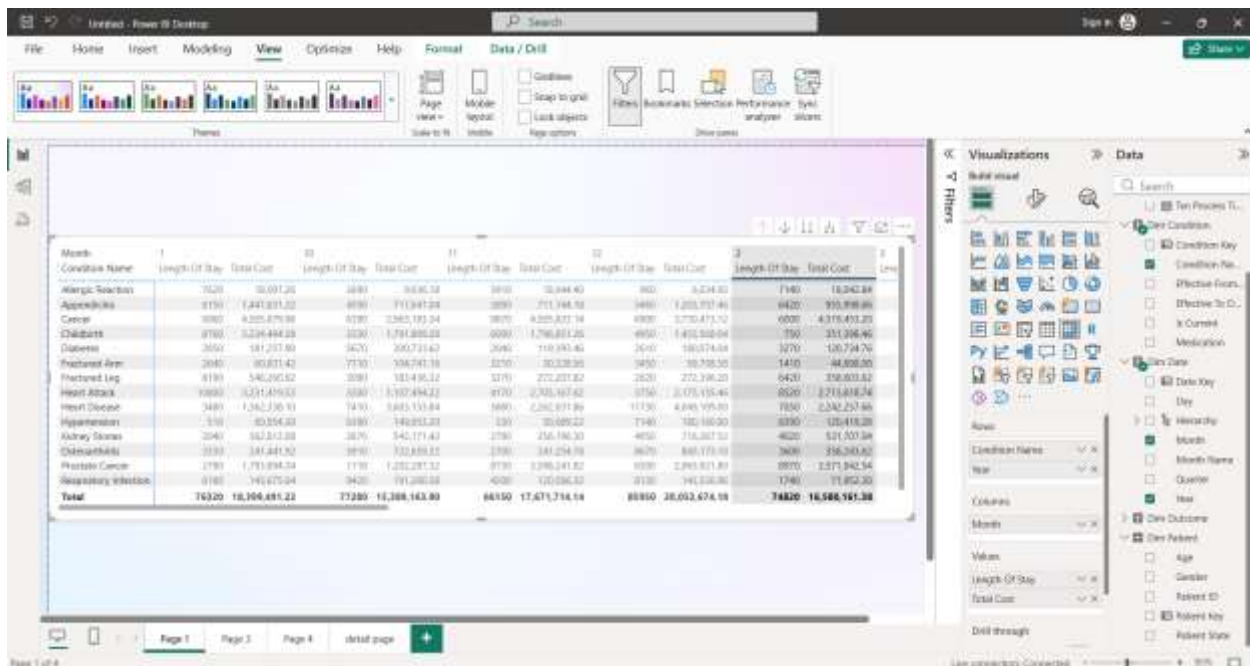


Figure 18 – matrix report

4.3 Report 2 — Slicers and Interactive Visuals

An interactive report was created using slicers and charts. Slicers allow users to filter data dynamically. Cascading filtering was implemented where selecting a year affects the available conditions. Charts provide clear insights into hospital data.

Steps-

1. Insert **Slicer visual**
 - Add field → Year
2. Insert another slicer
 - Add field → Condition Name
3. Add **Bar Chart**
 - Axis → Condition Name
 - Values → Length of Stay
4. Add **Column Chart**
 - Axis → Month
 - Values → Total Cost
5. Test filtering:
 - Select Year → check Condition updates

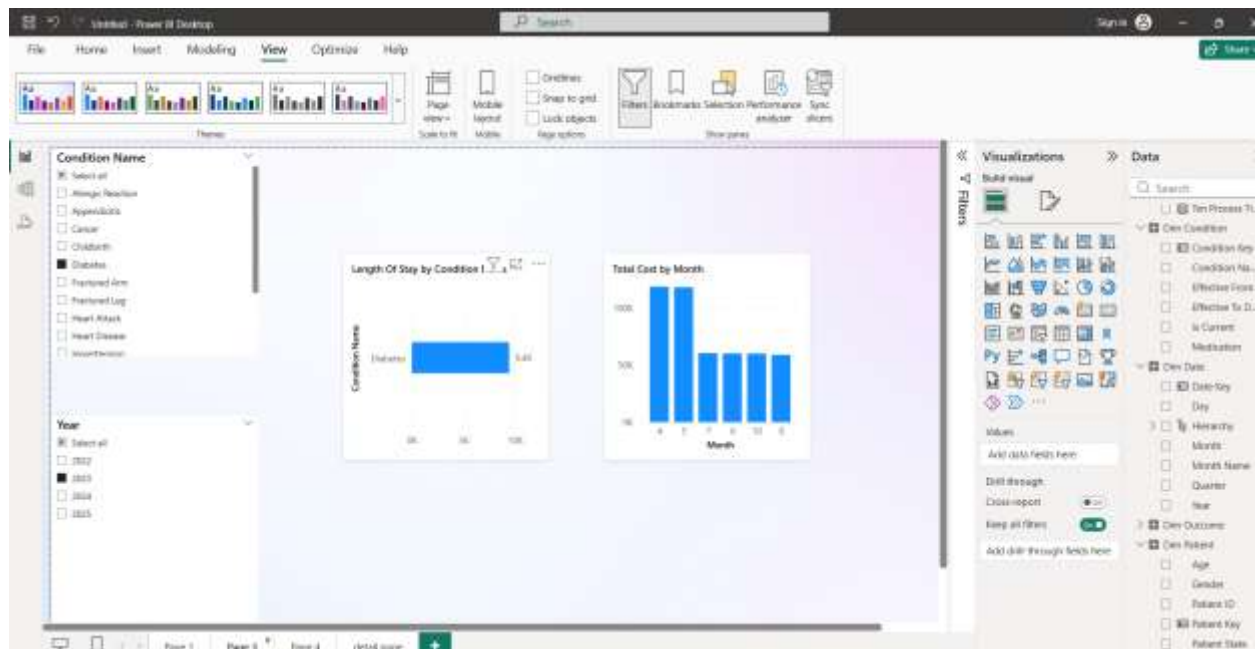


Figure 19- slicer report

4.4 Report 3 — Drill-Down Report

A drill-down report was created using a hierarchical structure. Users can explore data from higher levels (Year) to lower levels (Month). This improves analysis by providing detailed insights.

Steps-

1. Insert **Column Chart**
2. Add hierarchy:
 - Axis → Year → Quarter → Month
3. Add Values → Total Cost
4. Enable **Drill-down mode**
5. Click on chart to drill into data



Figure 20 – drill down report

4.5 Report 4 — Drill-Through Report

A drill-through report was created to view detailed information. Users can right-click a value and navigate to a separate page. This provides deeper insights into selected data

Steps-

1. Create New Page → rename as “Detail Page”
2. Click empty area
3. Add Drill-through field:
 - Condition Name
4. Insert Matrix visual
5. Add:
 - Rows → Condition Name
 - Values → Total Cost, Length of Stay, Satisfaction
6. Go back to main page
7. Right-click on a value → Drill-through → Detail Page

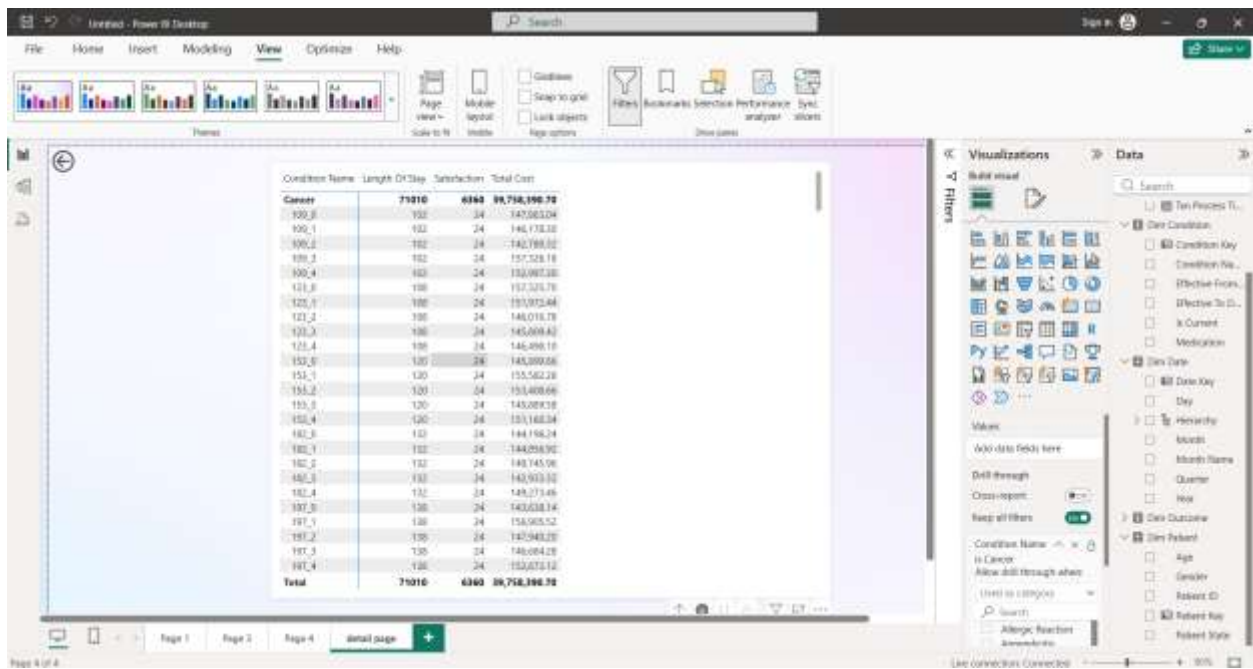
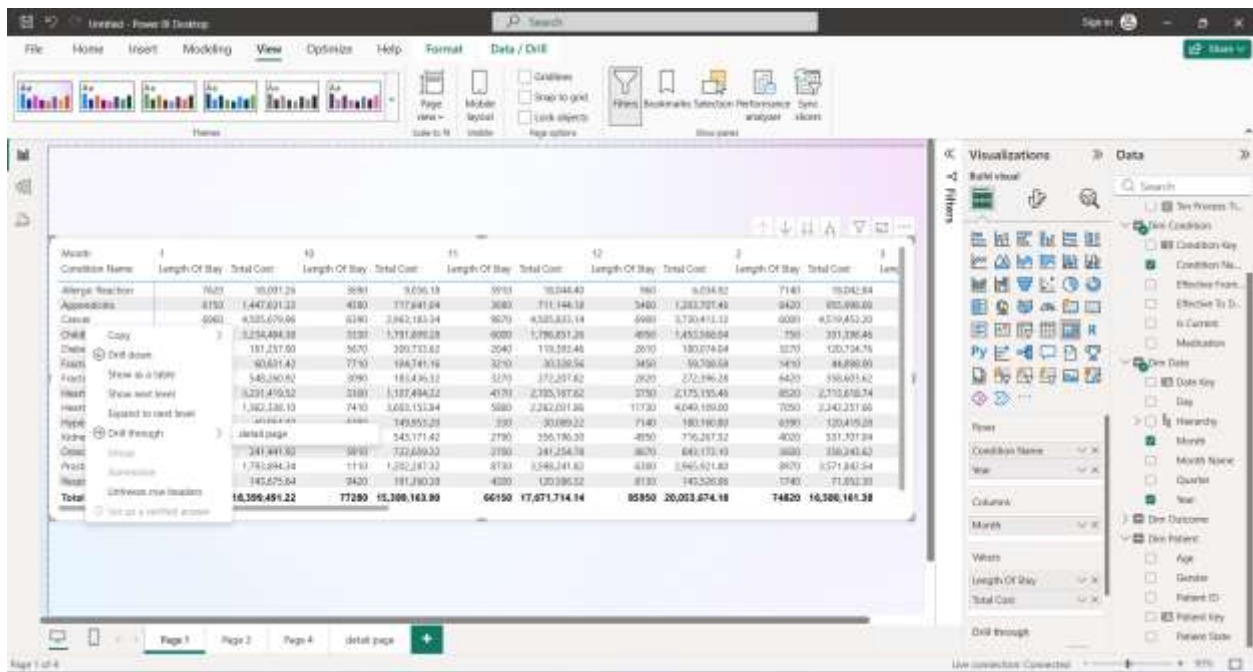


Figure 21-drill through

4.6 Visual Designs

Appropriate visuals were selected based on the type of data. Matrix was used for detailed analysis, bar and column charts for comparison, and line charts for trends. Slicers were used for interactive filtering, and the layout was designed for clarity and usability.

1. Selected appropriate visuals based on data type
2. Used Matrix for detailed tabular analysis
3. Used Bar and Column charts for comparison
4. Used Line chart for trend analysis
5. Added slicers for interactive filtering
6. Maintained clear layout and readability

Summary

Power BI reports were created using the SSAS cube to perform interactive data analysis. Features such as matrix visualization, slicers, drill-down, and drill-through were implemented. DAX was used to create additional calculations. These reports support better decision-making by providing clear and dynamic insights.